LEO ROULEAU

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EDUCATION

Polytechnique Montreal, *Bachelor's Degree in Software Engineering* 2025 - Present

- **Expected graduation:** December 2029, **Dean's List**

College André-Grasset, *College Diploma (D.E.C) in Computer Science and Mathematics* 2023 - 2025

EXPERIENCE

Matrox Video, *Incoming Software Engineering Intern* *Jan 2027 - Apr 2027*

- Confirmed C++ software development internship.

Google Summer of Code, Haiku OS, *Open-Source Contributor* May 2026 - Aug 2026

- Transformed the Devices preferences application into an interactive hardware manager using C++ and the native Haiku API framework.
- Extended the OS device management subsystem in C++ to expose active kernel driver modules, hardware attributes, and /dev filesystem paths to user space.
- Implemented comprehensive USB tree mapping, enabling inspection of interface descriptors and HID report formats.
- Added driver management controls, empowering users to safely blacklist device drivers directly from the GUI.

DRA Global, *Electrical Engineering Intern* *May 2026 - Aug 2026*

- Automated the generation of complex electrical and instrumentation drawings utilizing EPLAN Electric P8 and Excel scripting, reducing manual drafting time by over 90%, from multi-day tasks to under an hour.
- Developed custom VBA scripts to dynamically parse engineering data and automatically render schematic layouts.
- Collaborated with Senior and Principal Electrical engineers to perform technical drawing reviews, ensuring adherence to company standards and engineering quality benchmarks.
- Presented custom automation workflows to an international audience of 20+ engineers and senior management across the North American, South American and South African offices.

Service Télématique Étudiant de Polytechnique (STEP), *Treasurer (Previously: VP of Cybersecurity)* Sep 2025 - Present

- Developed automated virtual machine templates using Packer and Ansible on Proxmox VE, accelerating base environment deployment from roughly 1 hour to under 5 minutes while ensuring consistent provisioning.

EURO CAR WASH LTD, *Front-end Developer Intern* *Jun* 2025 - Aug 2025

- Developed and styled reusable React components to match the company's updated Figma design specifications.
- Ensured UI consistency by fixing layout issues and correcting design mismatches in TypeScript.

PROJECTS

C Bytecode Virtual Machine

- Engineered a stack-based bytecode VM in C, featuring a single-pass compiler and Pratt expression parser.
- Implemented a mark-and-sweep garbage collector and dynamic hash tables to handle memory management and object allocation safely.

Tree-Walk Interpreter

- Developed a complete tree-walk interpreter encompassing a custom Lexer, a recursive descent parser, and an abstract syntax tree evaluator in Java.
- Supported complex language features such as static scoping, control flow, functions, classes, and inheritance.

TECHNICAL SKILLS

Programming Languages: C/C++, Java, Python, JavaScript, TypeScript, VBA, Emacs Lisp

Tools & Environment: Linux, Git, Docker, Proxmox, Packer, Jira, QEMU, EPLAN Electric P8, Figma, Microsoft Suite